

Sisi Li

EDUCATION 408-480-6912 | susieli@uchicago.edu | www.linkedin.com/in/sisili144 | Mountain View, CA

University of Chicago | *M.S. in Data Science* | AI agents Chicago, IL | Sep 2024 - Mar 2026
Tianjin University | *M.S. in Computer Science* | Data Visualization Tianjin, China | Sep 2021 - Mar 2024
Tianjin University | *B.E. in Software Engineering* | GPA: 3.75 / 4 Tianjin, China | Sep 2017 - Jul 2021

WORKING EXPERIENCE

Samsung Research America | *AI Software Engineer Intern* Mountain View, CA | June 2025 - Dec 2025

- Developed AI search interface for Samsung Internet using **React** and **TypeScript**, implementing multimedia-aware incremental rendering and lazy loading, reducing first paint time by 38% and interaction latency by 45%
- Built browser automation extension using **Chrome DevTools Protocol** with persistent **WebSocket** connections and retry mechanisms, achieving 92%+ execution reliability across complex multi-step automation workflows
- Implemented **Django** backend orchestration system supporting AI-driven workflows with thread-based concurrent session management for parallel task execution, reducing median completion time by 65%
- Designed **MCP** server framework enabling AI agents to connect with Samsung services, delivered Naver Mail integration POC with PostgreSQL credential management and token authentication for automated email workflows

Oumiao | *Software Engineer (Part-time)* Tianjin, China | Jul 2022 - Mar 2024

- Applied **LoRA** and **PEFT** to fine-tune Llama-2-7B LLM model for context-aware QA tasks, integrating **RAG** pipeline to enhance contextual relevance, achieving a 16% improvement in coherence and a 7% enhancement in factuality
- Engineered the fine-tuned model into a Chrome extension with **Flask** backend, implementing **RESTful APIs** and **PostgreSQL** for context management and user data persistence, handling 8K+ daily queries with sub-200ms latency
- Established containerized deployment infrastructure on **AWS** using **Docker** on Linux, implementing auto-scaling and health monitoring that improved system reliability and enabled horizontal scaling to handle increased traffic loads
- Built an end-to-end **CI/CD** pipeline using GitHub Actions with automated code linting, testing, and reliable deployment, maintaining consistent code quality and reducing deployment time by 42%

Meituan (China's DoorDash + Groupon) | *Software Engineer Intern* Beijing, China | Jun 2020 - Sep 2020

- Built ticket validation system with dynamic QR code generation for attractions using **Vue**, reducing median server latency by 20% and improving entry processing for 500K+ daily users across 3K+ merchants; completed 20+ code reviews
- Implemented secure validation using digitally signed QR codes with auditable timestamps and hash-based deduplication, enabling O(1) lookups and achieving 99.9% accuracy while preventing duplicate or counterfeit redemptions
- Built a scalable, **full-stack** Chrome extension with **AWS Lambda-based** serverless backend to aggregate and index research findings. Enabled 100+ researchers to access information 21% faster, streamlining collaborative workflows

TWT Studio | *Software Engineer Intern* Tianjin, China | Jul 2018 - Apr 2020

- Built a React-based quiz platform with OAuth 2.0 SSO, auto-save, and offline support, serving 2K+ concurrent students; optimized **frontend** performance using Webpack code splitting and tree-shaking, cutting render time by 30%
- Collaborated **cross-functionally** with backend and product teams, leveraging Git-based **version control**, **code reviews**, and **object-oriented** design principles to deliver maintainable, scalable features in an Agile workflow

PUBLICATIONS

- **EvoVis: A Visual Analytics Method to Understand the Labeling Iterations in Data Programming** (Sisi Li, Guanzhong Liu, Tianxiang Wei, Shichao Jia, Jiawan Zhang) *IEEE Transactions on Visualization and Computer Graphics*
- **EddyVis: A visual system to analyze eddies** (Runpu Yue, Guan Li, Sisi Li, Suihua Shan) *Journal of Visualization*

PROJECTS

- **AI-Powered Recipe Management Platform:** Developed full-stack recipe platform with nutrition tracking, shopping list automation, and personalized recommendations using collaborative filtering algorithms, implemented **Redis** caching for 50K+ recipe queries with sub-100ms response time, deployed **Java Spring Boot** backend and React frontend on AWS

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, SQL, C/C++, Go, Rust, C#, HTML5, CSS

Frameworks: React, React Native, .NET, Angular, Next.js, Node.js, Spring Boot, Django, FastAPI, TailwindCSS, Vite, LangChain

Databases & Cloud: MySQL, PostgreSQL, NoSQL, MongoDB, Redis, Azure, Docker, Kubernetes, Terraform, Kafka, Snowflake

DevOps & Tools: Git, CI/CD, GraphQL, GCP, Jest, Bazel, PyTorch, TensorFlow, Microservices